Unit 1 (20 Marks)			
S. No.		**Marks**	**CO**
Q 1	Define Superintelligent AI. Explain with suitable example why it is considered hypothetical?	**4**	**CO1**
Q 2	List any four applications of AI in real-world industries.	**4**	**CO1**
Q 3	Define Machine Learning. How is it different from human intelligence?	**4**	**CO1**
Q 4	Discuss the functions of AI Agents with suitable examples.	**4**	**CO1**
Q 5	Compare Machine Learning and Deep Learning with suitable examples.	**4**	**CO1**
Unit 2 (20 Marks)			
Q 6	Explain Implication and Biconditional logical connectivity with truth conditions.	**4**	**CO2**
Q 7	Explain with suitable examples, all types of Quantifiers.	**4**	**CO2**
Q 8	Construct truth table for: $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$	**6**	**CO2**
Q 9	Prove using resolution method: KB1: $(p \rightarrow q)$ KB2: $(q \rightarrow r)$ KB3: p Prove that: r	**6**	**CO2**

Unit 1 (20 Marks)			
S. No.		**Marks**	**CO**
Q 1	Define Learning Agent and its components.	**4**	**CO1**
Q 2	State two advantages and two disadvantages of AI.	**4**	**CO1**
Q 3	Explain AI applications in Healthcare and Finance with examples.	**4**	**CO1**
Q 4	Define Machine Learning. Explain the types of ML with example.	**4**	**CO1**
Q 5	Explain, Perception → Decision → Action → Feedback, core cycle of agents.	**4**	**CO1**
Unit 2 (20 Marks)			
Q 6	Consider the following knowledge base: KB1: The humidity is high or the sky is cloudy. KB2: If the sky is cloudy then it will rain. KB3: If the humidity is high then it is hot. KB4: It is not hot. GOAL: It will rain. Use propositional logic and apply resolution method to prove that the goal is derivable from the given knowledge.	**8**	**CO2**
Q 7	Given the logic as: $\big((P \wedge Q) \rightarrow R\big) \rightarrow \big((P \wedge Q) \rightarrow (Q \rightarrow R)\big)$ Prove that LHS is logically equivalent to RHS	**8**	**CO2**
Q 8	Negate the quantified statement: $\forall x(P(x) \rightarrow Q(x))$	**4**	**CO2**

Unit 1 (20 Marks)			
S. No.		**Marks**	**CO**
Q 1	Differentiate between Narrow AI and General AI with examples.	**3**	**CO1**

Q 2	Describe various key components of an AI system.	**3**	**CO1**
Q 3	Define Artificial Intelligence. How is it different from human intelligence?	**2**	**CO1**
Q 4	Explain various types of AI Agents with suitable examples.	**4**	**CO1**
Q 5	Explain the major AI techniques: a) Machine Learning b) Deep Learning c) Natural Language Processing d) Computer Vision Support your answer with suitable real-world applications.	**2 + 2 +2 +2**	**CO1**
Unit 2 (20 Marks)			
Q 6	Explain the following logical connectives with truth conditions: (a) Conjunction (∧) (b) Implication (→)	**4**	**CO2**
Q 7	Define Predicate Logic? Differentiate it from Propositional Logic? Give suitable examples.	**4**	**CO2**
Q 8	Construct the truth table for: $(p \wedge q) \vee \neg p$	**6**	**CO2**
Q 9	Simplify the logical expression using logical laws: $\neg(p \vee q) \wedge (p \rightarrow q)$	**6**	**CO2**